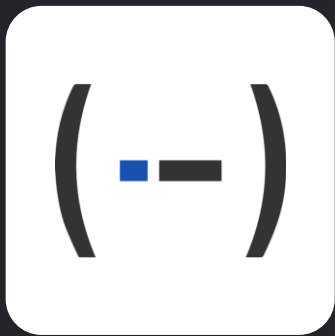




Case pattern

Build a chain of features once, against named inputs, then run it again per case -- each case with its own selections and values. Three features: Define case, Case, Close case.



What it does

'Apply to each' for the feature tree. Build a detail once for case A against names (#top, #bossH); each further case gives its own geometry and values, and the chain is run again for it. Nothing is moved by a transform, so the places may differ in shape.

Use it when

- The same feature chain applies to several places
- Each place has its own selections and dimensions
- The chain edits existing geometry (fillet, move face) as well as making new

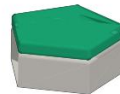
case A (Define case)
#bossH 15 mm, #edgeR 3



Case B
25 mm, 3

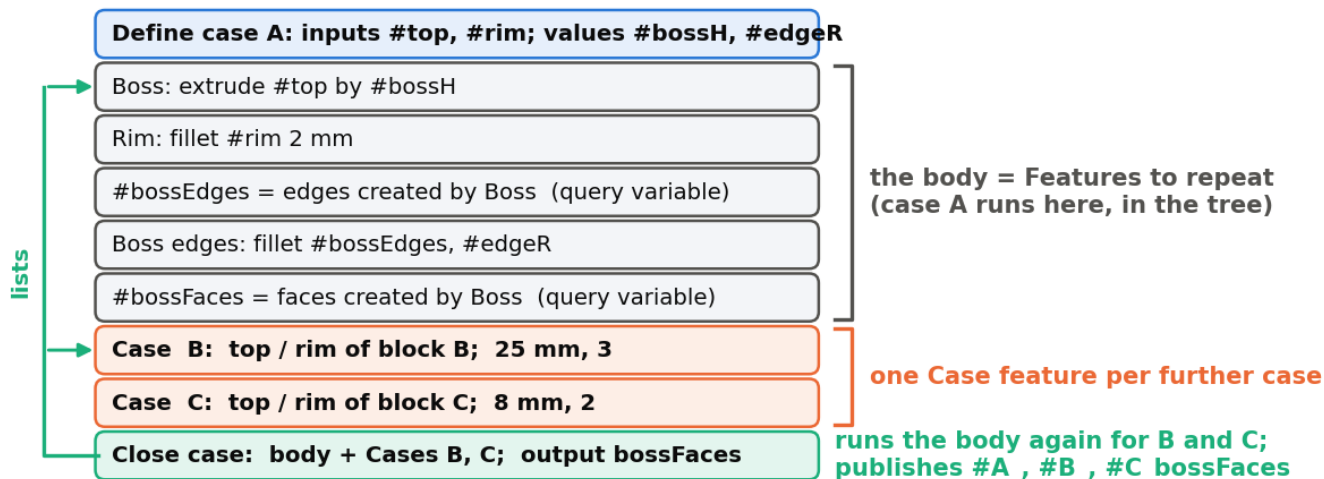


Case C
8 mm, 2



Test T1: one chain (boss, rim fillet, boss-edge fillet) built for case A and run again for B and C.

(--) Three features, like a function



Define case = the signature

Names the inputs and values, with case 1's selections. The features after it (the body) use those names.

Case = one call's arguments

One feature per further case: a selection per input, a value per value. Runs nothing itself.

Close case = runs the calls

Lists the body and the Cases, runs the body once per Case, publishes each case's outputs.

(--)

What the body may point at

An input (Define case)

#top from Define case;
each Case gives its own

per case

OK

From before the body, clicked

a tower face, a corner, a plane
(tests T9a, T13, T14)

same in every case

OK (v3)

Made by the body, via a query variable

#bossEdges = edges
created by Boss

remapped to each case's copy

OK

Made by the body, clicked

Boss's edges picked
in a later fillet

stays on case A's copy

ERROR (T3)

The one rule

Geometry the body itself made is
referenced through a query variable
(*'created by'* a body feature). A click stays
on case A's copy.

New in v3

Geometry from before the body may be
clicked directly (a side reference, an *'up
to'* face): every case uses the same one.

Values

Numbers and lengths are not geometry:
plain variables from anywhere earlier
work. Put one in Define case only if it
changes per case.



Define case

T1 Define case A: ?, ?; ? (...) ✓ ✗

1 Case 1 name A

2 Inputs ↑↓ CLEAR

> top ×

> rim ×

Add Input

3 Shared references

Add Reference

4 Values ↑↓ CLEAR

> bossH ×

> edgeR ×

Add Value

Final

#	Parameter	What it means
1	Case 1 name	The case the tree itself builds (letters, digits, _; starts with a letter).
2	Inputs	Up to 8 query variables that change per case: a name and case 1's selection.
3	Shared references	Optional: one selection for every case. Clicking the geometry in the body does the same; this is one place to change it.
4	Values	Up to 6 variables that change per case: name, type (Length, Angle, Area, Volume, Number, Integer, Text, Boolean), case 1's value.

Also sets #caseName and #caseIndex (1 for case 1); every case updates them.

(--)

Case

1

T1 Case B (25 mm, 3) ✓ ✗

Define case

T1 Define case A: ?, ?, ? (length... x

2

Update from Define case

3

Case name B

4

Input 1 #top

Selection

Face of T1 block B (9... x

⌚ (Q)

Input 2 #rim

Selection

Edge of T1 block B (9... x

⌚ (Q)

5

Value 1 #bossH (length)

Value 25 mm

Value 2 #edgeR (number)

Value 3

Final

#	Parameter	What it means
1	Define case	The Define case this case belongs to. Picking it lays out the slots.
2	Update from Define case	Lays the slots out again after inputs or values changed in the Define case; selections and values are kept by name.
3	Case name	Unique among the Define case's cases; also the output prefix (#B_...).
4	Input slots	A selection for each input (labelled #top, #rim ...).
5	Value slots	A value for each value. New slots start with case 1's value (a copy, not a link).

Place each Case after the body and before the Close case that runs it. A missing selection or value is an error on the Case.



Close case

1

2

3

4

5

6

7

T1 Close case B, C -> Bo...

✓

✗

Define case

T1 Define case A: ?, ?; ? (lengt... ✕

Features to repeat

T1 Boss: extrude ? ? new ✕

T1 Rim: fillet ? 2 mm (outside ... ✕

T1 ? = edges created by Boss ... ✕

T1 Boss edges: fillet 2.2 mm ✕

Cases

T1 Case B (25 mm, 3) ✕

T1 Case C (8 mm, 2) ✕

Outputs

CLEAR

> bossFaces ✕

Add Output

> Keep

☒ Name parts after outputs

☐ Print bindings

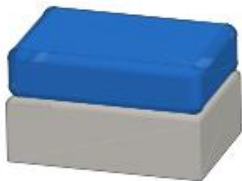
Final

#	Parameter	What it means
1	Define case	The Define case this body belongs to.
2	Features to repeat	The body's features, in tree order.
3	Cases	The Case features to run. May be empty: then it only publishes case 1's outputs.
4	Outputs	Per output: a Name and the query variable the body sets (its name as plain text, no #). Evaluate on use / Track downstream changes as in Extract variables.
5	Keep	Which new bodies each case keeps: parts, surfaces, curves and points, mate connectors, planes on; sketches off.
6	Name parts after outputs	On: a part an output points to is named like the output's variable (B_rib).
7	Print bindings	Print each case's selections and values to the notices.

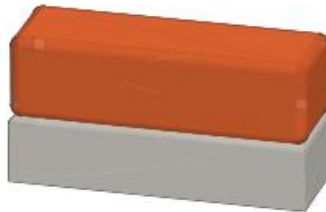


Example: bosses on three blocks (T1)

case A (Define case)
#bossH 15 mm, #edgeR 3



Case B
25 mm, 3



Case C
8 mm, 2



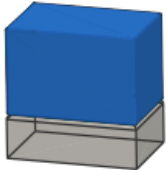
Define case A: #top, #rim; #bossH, #edgeR. Body: Boss (extrude #top by #bossH), Rim fillet on #rim, #bossEdges = edges created by Boss, fillet #bossEdges at #edgeR, #bossFaces = faces created by Boss.

Case B (25 mm, 3) and Case C (8 mm, 2); one Close case with output bossFaces. Result: three bosses at their own heights, every rim and boss edge filleted; parts A_bossFaces, B_bossFaces, C_bossFaces and the query variables #A_bossFaces, #B_bossFaces, #C_bossFaces.

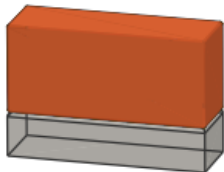


Example: a reference clicked from before the body (T14)

case A



Case B



**tower, before the body:
its top is clicked in the extrude**



The boss is extruded 'up to' the top of an outside tower, clicked directly in the extrude; the body also fillets the case's block rim and the boss edges.

Case B's boss stops at the same tower top as case A's (z 20..60); B's rim and boss edges are filleted. All three kinds of reference in one case: clicked outside geometry, an edit of geometry from before the body, a query variable made in the body. In v2 the clicked face was lost.

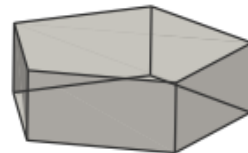
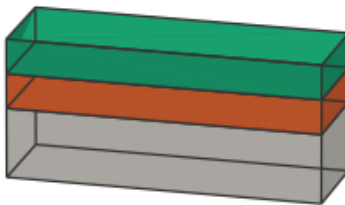
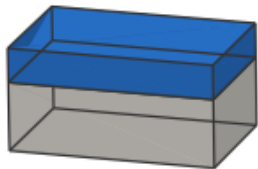


Example: closing again (T7)

B_stud (first Close case)

A_stud (case 1)

D_stud on top of B's stud (second Close case)



A first Close case runs Case B (a stud on block B). Case D selects the top of B's stud; a second Close case, on the same Define case with the body picked again, runs it.

D's stud sits on B's (z 30..40); #A_stud, #B_stud and #D_stud each hold one body. Use this when a case must select geometry an earlier case made.



Outputs and messages

Variables

Key	What it holds
#<case>_<output>	One query variable per output per case, case 1 included (#A_bossFaces, #B_bossFaces ...). Fixed when the case closes unless Evaluate on use.
inputs, values	Define case binds them to case 1; after the Close case later features see case 1 again.
#caseName, #caseIndex	The running case's name and number (case 1 = 1); back to case 1 after the Close case.
other body variables	Keep the LAST case's value after the Close case.

Messages

	Message / what to do
Warning	Not built: B: repeated feature 3 failed (...) The other cases are still built; Onshape's own reason is quoted.
Error	No case was built. Every listed case failed.
Error	#h has no value: click Update from Define case. On a Case: the Define case changed (T11 shows the error).
Info	#B_x is not case B's geometry: make #x a query variable 'created by' a repeated feature The output was not remade per case.
Info	Sketches are re-solved per case ... Constraints to the origin / default planes keep case 1's position.



Tips and limits



Few, short local names

Query variables made in the body are locals: name them by role (`_body`, `_edges`, `_out`) and reuse them in every template. A name can never be deleted from the list.



Outputs only for what is used later

Each output makes one name per case. Type output names as plain text: a leading `#` turns the field into a reference.



Switch features per case

A Boolean value suppresses a feature by expression (`#pin`), re-checked per case (T6). Numeric fields take expressions: `#flag ? 3 mm : 0 mm`.



Order and undo

Cases run in order and see earlier results; a failed case cannot undo its edits to existing geometry (only its new bodies are removed).



Tested edits

Fillet, Move face, Move face+ on geometry from before the body; chamfer, shell, draft, booleans into existing parts are untested. No sheet metal or derived features.



Cost and dialog

Every rebuild runs the body once per case. In a NEW Case the grey slot names stay blank until Update from Define case is pressed once.



Where it lives

Document	Case_Pattern (2099413dd91f34578b385892)
Tab	case_pattern (Define case, Case, Close case)
Tests	Case pattern tests studio: T1-T14 (T3, T11 must error), 53/53 checks pass 2026-09-29 -- devtools/onshape/build_case_pattern_tests.py, check_case_pattern_tests.py
Explainer	docs/explainers/case_pattern/case_pattern_explained.md
Design	case_pattern/DESIGN.md sections 11-12; corrections log 41, 50, 60
From v2	Case pattern features are gone: each case row becomes a Case feature listed in the Close case; outputs are typed as query-variable names.